Design and Validation of a Novel Human–Machine Interface System Based on Multisource Data Fusion

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***Abstract*— Conventional human–machine interface (HMI) systems generally take surface electromyography (sEMG) as the only signal source, which often faces the problem of multichannel crosstalk. To enhance HMI performance, this study introduces a novel three-degree-of-freedom (3-DOF) HMI system for surgical robot control, which integrates sEMG signals and inertial measurement unit (IMU) signals. A cost-effective wearable HMI hardware system is designed to address the high price and poor portability of existing commercial equipment, which also has a higher signal-to-noise ratio (SNR) of 35.9 dB. Based on the proposed simultaneous and proportional control (SPC) strategy, ten subjects were recruited to conduct an online experiment involving four upper limb postures to assess the real-time performance of the HMI system. In addition, six able-bodied participants participated in a validation experiment to statistically verify and rigorously evaluate the implemented HMI system. The experimental results demonstrate that our HMI system can accurately decode the angles of the wrist and metacarpophalangeal (MCP) joints in both single-DOF and 3-DOF movements, achieving stable control and robust online performance. Furthermore, the proposed HMI system was applied to a flexible endoscopic robot to realize remote SPC, and the results indicate that it can perform well in the practical application. The current study proves the feasibility and practicability of multisource data fusion for HMI, providing a new direction.**

***Index Terms*— Electromyography, human–machine interface (HMI), inertial measurement unit (IMU), multisource data fusion, musculoskeletal model, simultaneous and proportional control (SPC), surgical robot control.**

NOMENCLATURE

ADC	Analog-to-digital converter.
CNT	Counter.
DOF	Degree of freedom.
GUI	Graphical user interface.
HMI	Human–machine interface.
IIC	Interintegrated circuit.
IMU	Inertial measurement unit.
MCP	Metacarpophalangeal.
MCU	Microcontroller unit.
MVC	Maximum voluntary contractions.
NRMSE	Normalized root mean square error.
r	Pearson correlation coefficient.
rms	Root mean square.
sEMG	Surface electromyography.
SNR	Signal-to-noise ratio.
SPC	Simultaneous and proportional control.
UART	Universal asynchronous receiver and transmitter.

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I. INTRODUCTION

THE HMI is one of the important ways to realize the communication between human and machine. It can "translate" the movement intention of human to the machine to realize the target action and transfer the situation of the machine end-effector back to the human, so as to update the control instructions accordingly. In recent years, HMI has become a research hotspot in the fields of biomedicine, aerospace, and even virtual reality [1].

With the development of general surgery toward minimally invasive surgery, the application of surgical robots has become more and more widespread, making great contribution to the reduction of patient pains and hospitalization time [2]. In robotic-assisted minimally invasive surgeries, the surgeon generally controls the robotic instruments with the assistance of an intelligent control system, which enables the completion of many complex operations in a more precise, flexible, and controllable manner, thus reducing the overall complexity of the task. However, surgical robots are featured by limited operation dexterity, sensory feedback, and visual feedback, since current control interfaces (e.g., keyboard, mouse, and joystick) are bulky and difficult to use [3], which does not conform to the principle of intuitive control.

The HMIs based on sEMG, which can reflect muscle activation and movement patterns [4], provide a new direction for solving this problem. Traditional sEMG-based HMIs mostly adopt amplitude-based direct control, which requires nonintuitive switching commands [5]. With the development of pattern recognition algorithms, the classification accuracy

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of healthy individuals has exceeded 95% [6], [7], [8], [9]. Shin et al. [10] used dynamic gestures and inferred muscle force from sEMG signals to activate/deactivate the sEMG-based HMI system, thereby achieving the control of a 6-DOF robot with a 1-DOF gripper. However, pattern recognition can only classify sEMG signals into several discrete joint angles, which is inconsistent with the natural movements of human limbs [4].

The SPC algorithms based on sEMG signals have also been widely studied. Artificial neural networks [11], linear regression [12], nonnegative matrix factorization [13], and other advanced data-driven algorithms have been used to develop models for SPC [14]. However, these algorithms are data-driven black-box methods that rely on the digital mapping between input and output signals, requiring large amounts of training data and being unable to accurately predict new movements not included in the training data [15]. Researchers have applied prior knowledge of human physiological structures to musculoskeletal models [16]. Wrist flexion/extension, MCP flexion/extension, and wrist pronation/supination are the main DOFs of hand movement. However, challenges remain for muscles with a relatively low SNR, such as the supinator [17]. In addition, crosstalk in sEMG signals limits the accuracy of decoding multi-DOF movements. Therefore, HMIs that use sEMG as the sole input cannot fully achieve SPC for multi-DOF surgical robots.

In practical applications, it is necessary to use multisource inputs for the control of complex devices. In recent years, HMI with multisource data fusion has received much attention from researchers, and more and more fusion schemes have been proposed. Previous studies have demonstrated that ultrasound, near-infrared spectroscopy, mechanomyography, IMU, and electroencephalogram signals can provide auxiliary information for HMI [18], [19], [20], [21], [22], [23], [24]. Among them, the small-sized, low-cost IMU provides sufficient movement data for HMI, which requires no direct contact with the skin and has a higher SNR than sEMG. Therefore, the joint application of IMU and sEMG signals is the most studied in the literature and has proved to be feasible for data fusion [25], [26], [27], [28], [29], [30].

The integration of IMU and sEMG signals has seen quite a few applications for multi-DOF robotic control. Gowtham et al. [31] collected single-channel sEMG and IMU signals to control a 5-DOF robotic arm. Similarly, Morais et al. [32] used the Myo armband system to capture sEMG and IMU data, achieving control of operations for the PeopleBot robot via the robot operating system platform. Fall et al. [33] used IMU to read head orientation angles for proportional joystick command emulation and used sEMG to measure muscle activation for activating buttons B1 and B2.

These studies generally used IMU data for position or posture control and sEMG data for hand command control. This is because wrist and MCP joint movements typically occur in low-speed or small-amplitude scenarios, where IMU data lack the sensitivity to capture subtle changes [34] and can result in cumulative errors (drift) during use [35]. In fact, the current studies primarily implemented either direct control [31], [33] or discrete control methods based on pattern recognition [32], mapping predefined muscle activation or gestures to robot switching commands or actions [36]. While straightforward, these methods limit robotic motion capabilities to a few predefined commands or actions and lack intuitive control, requiring users to undergo extensive training.

In contrast, our HMI system provides continuous and intuitive control by decoding joint angles, allowing the robot to reach specific angles or execute continuous commands. Unlike previous studies that focused on binary or discrete control states (e.g., "on" and "off"), our approach enables more flexible and refined interactions with the robot. This advancement significantly improves the adaptability and functionality of the controlled device.

Furthermore, to enable the synchronized acquisition of sEMG and IMU data, our study designs and systematically validates a low-cost, wearable HMI hardware system. Our work also explores the potential of integrating sEMG and IMU data for remote SPC of a flexible endoscopic robot.

Specifically, the contributions of our study to the state-of-the-art in the field are concluded as follows.

1) *The Novelty of the HMI Hardware System:* Our research develops a low-cost, wearable HMI hardware system that integrates sEMG and IMU signals. Unlike the existing commercial devices (e.g., DELSYS Trigno or Myo armband), our system offers higher flexibility in signal acquisition and processing, making it both cost-effective and customizable for various applications.
2) *Advancements in Multi-DOF Control:* With an improved musculoskeletal model, the system can decode 2 DOFs of wrist and MCP joint movements from sEMG signals, while incorporating IMU data to extract a third DOF for wrist pronation/supination.
3) *Real-Time Performance and Practical Applicability:* The system underwent extensive validation through online experiment and HMI validation experiment. The SPC strategy of the system was, for the first time, successfully applied to the remote operation of a flexible endoscopic robot.

The rest of this article is organized as follows. Section II introduces the hardware and software design of the sensor system, along with the GUI and control algorithms of the HMI. Section III covers evaluation methods and experimental paradigms. Section IV presents the experimental results. Section V discusses the findings, and Section VI concludes this study.

II. Method

A. Hardware Designs

The hardware of the novel HMI system based on multisource fusion mainly refers to the sensor system, which consists of the circuit and structural component.

1) Overall Design of the Circuit: The circuit of the sensor system primarily includes the main controller, the sEMG acquisition module, the IMU module, the wireless communication module, and the power management module. The sEMG acquisition module, in particular, includes the sEMG electrode, a signal conditioning module, and a boost converter module. After the sEMG electrode collects the signals, the

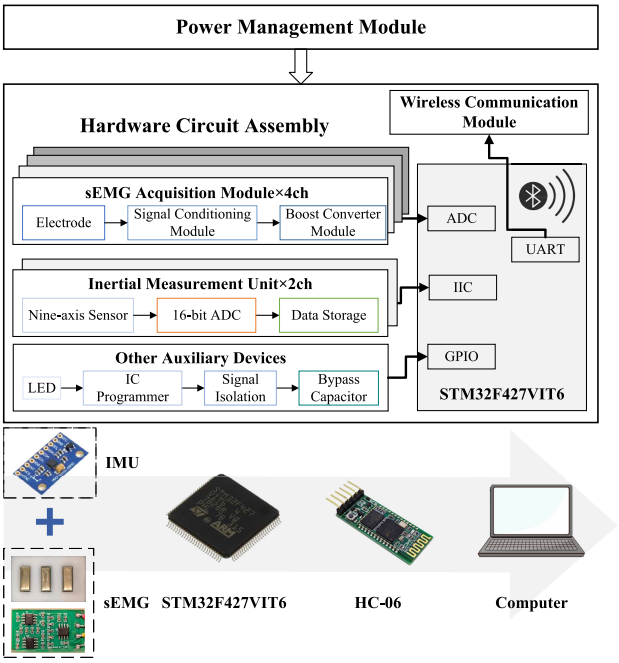


Fig. 1. Overall circuit design of the HMI system based on the fusion of sEMG and IMU, which primarily includes the main controller, the sEMG acquisition module, the IMU module, the wireless communication module, and the power management module.

signal conditioning module applies second-order Butterworth high-pass and low-pass filters at 10 and 500 Hz, respectively, with a gain of 500 V/V and a boosted voltage of 1.5 V. The 12-bit ADC of the MCU (STM32F427VIT6, STMicroelectronics, Geneva) is used to achieve 1-kHz analog-to-digital conversion, with a voltage range of 0–3.3 V and a resolution of 0.8057 mV. The IMU mainly consists of a nine-axis sensor (MPU9250, TDK InvenSense, CA) and a 16-bit internal ADC, and the raw data at the chip registers are read through the IIC of the MCU. The wireless communication module is based on Bluetooth 2.0 protocol, and the Bluetooth chip (HC-06, HC TECH, Guangzhou) transmits the digital signals to the computer with the UART. Fig. 1 presents the overall circuit design of the HMI system based on the fusion of sEMG and IMU, and the detailed design of the key modules will be introduced in Sections II-A2–II-A4.

2) Design of the Signal Conditioning Circuit for sEMG Acquisition Module: The sEMG signals are weak [37], ac low-frequency [38], overdriven [39], and high-impedance. Based on these characteristics, the sEMG acquisition module is designed to minimize ambient noise and capture sEMG signals with a higher SNR. The sEMG acquisition process of the HMI system is indicated in Fig. 2(a). The raw sEMG signals are processed by a differential amplifier (AD8422, Analog Devices, MA), a high-pass filter (AD8638, Analog Devices, MA), a low-pass filter (AD8638), and the boost converter module (AD8638), converted into the range of the ADC, and finally converted into digital signals for processing by ADC. The schematic of the signal conditioning circuit is shown in Fig. 2(b).

TABLE I
VOLTAGE REQUIREMENTS OF HMI HARDWARE CIRCUIT CHIPS

Chip	Voltage requirement	Supply voltage
STM32F427VIT6	1.7-3.6V	3.3V
MPU9250	2.4-3.6V	3.3V
HC-06	3.0-3.6V	3.3V
AD8422	±1.8V to ±18 V	±5V
AD8638	±3.7V to ±18 V	±5V

Here, a gain multiplier of 200 is chosen to avoid distortion of the sEMG signals in the amplification process. The second-order high-pass filter and the second-order low-pass filter together form a fourth-order Butterworth bandpass filter, which has a passband of 10–500 Hz. In addition, a boost converter circuit, implemented using an adder, sets the boost voltage to 1.5 V.

Notably, the ground connected to the sEMG acquisition module is entirely analog ground. To prevent fluctuations of voltage in the digital ground from affecting the analog signal acquisition, a 600-Ω resistor is placed between the analog ground and the digital ground for signal isolation.

3) Design of the Power Management Module: The HMI hardware circuit designed above involves different types of chips, which have various voltage requirements, although we have taken the consistent supply voltage into account as much as possible in the chip selection. Table I presents the voltage requirement and supply voltage for each chip.

In this study, two 5-V rechargeable Li-ion batteries (each 5600 mAh) are connected in series to power the sensor system, with the midpoint voltage used as the reference ground. A linear regulator (AMS1117-3.3, UMW, Shenzhen) is selected as the power supply for the controller, the IMU, and the Bluetooth module, and the power supply interfaces of each chip are equipped with filtering capacitors to prevent the voltage from fluctuating.

4) Hardware Circuit Assembly: According to the above design scheme, we design a kind of two-layer printed circuit board (PCB), with reasonable layout and wiring of its components, as shown in Fig. 2(c).

For the circuit of our sensor system, altogether two PCBs are processed, as the main controller board and the sEMG sensor board. Moreover, an IMU module (MPU9250) and a Bluetooth module (HC-06) are purchased. As shown in Fig. 2(c), the main controller consists of the MCU module (H), power supply module (G), boost converter circuit (E), and interfaces for the sEMG sensor board (B) and IMU module (C).

5) Wearability Design: The ultimate goal of our HMI system is to achieve robust multi-DOF control. To meet this requirement, the sEMG electrodes need to be affixed to the specific muscle belly, and the IMU needs to be affixed to the corresponding position according to the decoding requirement. Instead of the wearability of specific sensing units, here we mainly focus on the wearability of the entire HMI system, namely, the controller and its connection with the sensing units.

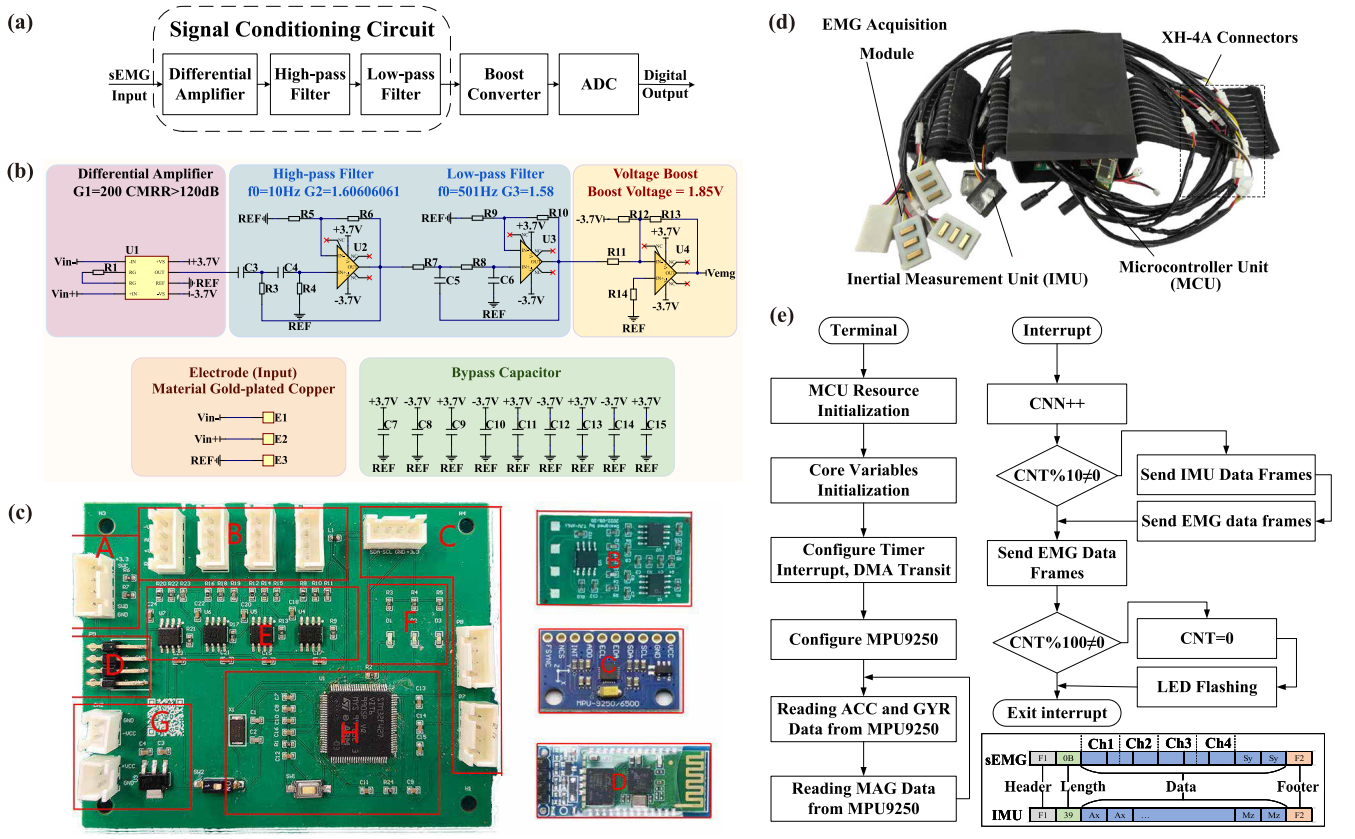


Fig. 2. Hardware and software design for the sensor system. (a) sEMG acquisition process. (b) Signal conditioning module. (c) PCB, the hardware circuit (A. Programmer interface. B. sEMG sensor and its interface. C. IMU and its interface. D. Bluetooth module and its interface. E. Boost converter circuit. F. Signal indicator lights. G. Power management module. H. Microcontroller module). (d) Connection diagram of the sensor system. (e) Program flowchart of the MCU.

As mentioned, since the battery pack is large, a special platform is set on the main controller case to prevent the batteries from being bumped. Moreover, XH-4A connectors are set on the main controller to connect the sensing units (IMU and sEMG) to the main controller. This design extends the length of the connecting wires and simplifies component replacement. The main controller case is equipped with a self-adhesive elastic bandage, allowing the user to fix the sensor system at the waist. Fig. 2(d) illustrates the connection diagram of the complete sensor system.

B. Software Design

The software design of our HMI system consists of embedded programming for the MCU and GUI development for the computer interface. A clear and feature-rich GUI can facilitate the use of the whole HMI system, thus laying a good foundation for the commercialization and practical promotion of the system.

1) Embedded Programming: The design framework for the embedded program of the MCU is illustrated in Fig. 2(e). First, the main clock frequency of the MCU is set to the highest value of 180 MHz to achieve the fastest processing speed. Then the on-chip resources are initialized, mainly including IIC, UART, ADC, and timer. The initialization configuration of each resource is as follows.

1) Three IIC interfaces are enabled, and the IIC communication is configured as a fast mode with a clock frequency of 400 kHz.
2) One UART interface is enabled, the baud is set to 921 600 bit/s, the data length is 8 bits, no parity bit, and the length of the stop bit is 1 bit.
3) One timer is enabled, a 180× division frequency is adopted, the autofill register is set to 999, and the counting mode takes downward counting, so that the 180-MHz main clock frequency generates a timer interrupt every 1 ms.
4) Five ADC channels are enabled in multichannel (scan) conversion mode, allowing the voltages of the five conversion channels to be sampled continuously without occupying the resources of the MCU. The precision of each channel is set to 12 bits, the sampling time is set to 112 main clock frequency cycles, the total conversion time of the five channels is 620 cycles, and the theoretical sampling time of all the channels is about 3.44 μs, which meets the sampling frequency requirement of sEMG signals.

The data from the sEMG and IMU, as well as the CNT, are called at timer interrupts, so they need to be designed as global variables. Furthermore, we enable the timer interrupt function and transit the sampled data of the five channels into the predefined global variables via direct memory access.

Before reading the data from MPU9250, it is necessary to initialize its configuration. According to the manual of the chip and the requirements of subsequent experiments, we assign values to its configuration registers through IIC after power-on for 300 ms and configure its sampling frequency as 100 Hz, sampling precision as 16 bits, gyroscope range as ±2000 deg/s, acceleration range as ±2 g, and magnetometer range as ±4800 T. Since the MPU9250 is a composite chip consisting of MPU6500 and AK8975, the accelerometer and gyroscope share the same address in IIC communication, while the magnetometer has a separate address.

According to the configuration of the initialized part of the MCU, the timer triggering interval is 1 ms. The sampling frequency of the sEMG and synchronization signals is configured to 1000 Hz, while that of IMU is set to 100 Hz. Therefore, sEMG data frames are sent at each interrupt, while the IMU data frames are sent only when the interrupt count is divisible by 10. Each sEMG data frame is 11 Bytes in total, including 1 Byte of header, 1 Byte of footer, 1 Byte of data length tag, and 8 Bytes of data. Among them, the data portion is used to store the four-channel sEMG signals and one-channel synchronization signals. The sEMG signals of each channel are 1.5 Bytes, and thus, the sEMG signals of four channels occupy a total of 6 Bytes. The last 2 Bytes of the data only store the synchronization signals. Each data frame of IMU is 57 Bytes in total, which includes 1 Byte of header, 1 Byte of footer, 1 Byte of data length tag, and 54 Bytes of data. Among them, the data portion stores the data of the three-axis accelerometers, gyroscopes, and magnetometers of the three-channel IMU, and each of the data is 2 Bytes. Since the data type of the CNT is defined as uint16_t in the initialized core variables, the CNT is zeroed when the data exceed the upper limit. In addition, the CNT is zeroed when it reaches 1000, thus to avoid fluctuations in the data transmission frequency when the upper limit is exceeded.

2) Design of Offline Acquisition GUI: The raw data acquired by our sensor system are transmitted to the computer through the Bluetooth module. However, the raw data are not intuitive, making it difficult for users to immediately assess signal quality during acquisition and requiring complex operations for processing. A user-friendly GUI addresses these issues, significantly enhancing usability.

Based on the characteristics and applications of the sensor system, a GUI for offline data acquisition is designed based on MATLAB (MathWorks, Inc., MA). The functions of it mainly include information display, data saving and reading, and experimental paradigm assistance. It can be compiled into an executable program without the limitations of MATLAB, providing a convenient environment for the entire sensor system. Fig. 3(a) illustrates the specific layout of the GUI.

The raw sensor data transmitted through the Bluetooth module consist of two different data frames. We distinguish between consecutive data frames according to the frame header and footer, and between sEMG data frames and IMU data frames according to the data length label.

In the signal display module, the five-channel signals contained in the sEMG data frame are displayed graphically in real-time, so that the user can fine-tune the position of the sEMG electrodes according to the sEMG graphs, thus to obtain a signal source as independent as possible. The IMU data frame contains 27 channels from three IMUs, sampled at a lower frequency than sEMG and displayed in a table on the right.

In the experiment assistance module, we provide adjustable acquisition duration, upper limb postures, movement type, and experimental group for experimental data saving and create a separate folder for each subject to facilitate the subsequent processing of the data.

3) Design of Online Experiment GUI: The online experiment GUI is designed to read the data acquired by the sensor system, preprocess the signals, decode joint angles, and map them to the position and size of a virtual cursor. Before the online experiment of our HMI system, the sEMG electrodes should be attached to the bellies of the corresponding muscles to minimize the crosstalk between different channels, and then the sEMG signals under the MVC should be collected to normalize the subsequent sEMG signals. During the experiment, the target and the predicted joint angles should be intuitively displayed in the virtual cursor control module in real-time. At the end of the experiment, it is required to record whether the task is completed or not. If the task is completed, it is required to calculate the completion time, the path efficiency, and the number of overshoots. Based on these requirements, an online interface is designed using MATLAB, as shown in Fig. 3(b). The GUI mainly contains the following modules.

1) *Signal Display Module:* This module can display the four-channel sEMG signals in real-time, which facilitates the placement of the sEMG electrodes prior to the experiment, allowing the placement scheme that minimizes the crosstalk as much as possible.
2) *Model Calibration Module:* With this module, we can acquire sEMG-MVC signals from four muscles, which are used to normalize subsequent sEMG signals. The user can also manually adjust the linear gain to scale the input sEMG signals and the output joint angles.
3) *Virtual Cursor Control Module:* This module represents 3-DOF joint angles using cursor movements (up, down, left, right) and size variations. A black cursor represents the predicted joint angles in real time, while a gray cursor represents the target angles.
4) *Evaluation Metric Module:* This module can calculate the completion time, the path efficiency, and the number of overshoots in real-time after each goal is completed.

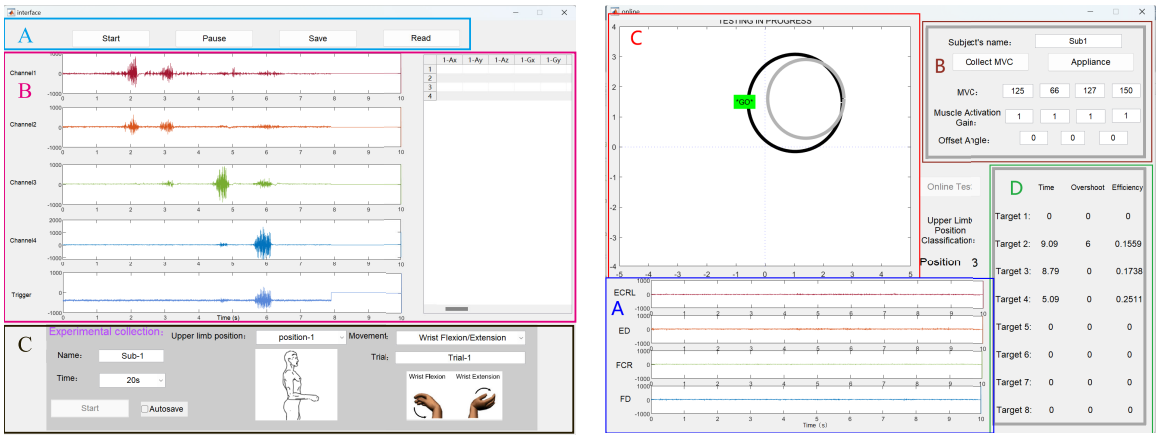
C. Model and Algorithmic

In this study, we propose a 3-DOF motion control strategy for the wrist and MCP joints based on the integration of sEMG and IMU signals. Specifically, four-channel sEMG signals are fed into an improved musculoskeletal model to control the flexion and extension of the wrist and MCP joints. Since it is challenging to use sEMG signals alone for controlling the robotic system, we incorporate an IMU to capture forearm movements and use the recorded data to control wrist pronation and supination.

For upper limb position classification, we use 18-channel data streams collected from IMUs mounted on the upper

TABLE II
SUMMARY TABLE OF THE DESIGN METHODS

Design	Module	Function	Method
Hardware designs	sEMG signal acquisition and preprocessing module	EMG signal acquisition, differential amplification, analog filter.	Copper electrode, AD8422, AD8638
	Inertial measurement unit (IMU) module	Sensing limb movement.	MPU9250
	Bluetooth module	Microcontroller unit (MCU) communicates and transmits data to the computer via Bluetooth module.	HC-06
	MCU module	Control, signal indicator, programmer, voltage boost.	STM32F427VIT6
	Power management module	Supply ±5V, 3.3V	Two 5 V rechargeable Li-ion batteries, AMS1117-3.3
Software designs	Embedded programming	On-chip resource allocation, analog-to-digital conversion, data frame fusion, IIC communication between MCU and IMU, UART communication between MCU and Bluetooth module.	KEIL
	Design of offline acquisition graphical user interface (GUI)	Data saving and reading module, signal display module, experiment assistance module.	MATLAB
	Design of online experiment GUI	Signal display module, model calibration module, virtual cursor control module, evaluation metric module.	MATLAB
Wearability designs	Case design	Prevents bumping, securing the sensor system to the waist.	3D printed hard case, self-adhesive elastic bandage.
	Connection design	Increase the length of connecting wires, Easy replacement of parts.	XH-4A connectors
Model and algorithmic	Musculoskeletal model based on sEMG signals	Decoding four-channel sEMG signals to predict joint angles for wrist flexion/extension and MCP flexion/extension.	MATLAB
	Linear regression based on roll angle	Decoding one-channel IMU data to predict joint angles for wrist pronation/supinatio.	MATLAB
	Linear discriminant analysis classifier	Upper limb position classification.	MATLAB



(a) (b)

Fig. 3. HMI offline and online GUIs. (a) HMI offline acquisition GUI: A. Data saving and reading module. B. Signal display module. C. Experiment assistance module. (b) HMI online experiment GUI: A. Signal display module. B. Model calibration module. C. Virtual cursor control module. D. Evaluation metric module.

arm and forearm. Each IMU provides data from a three-axis accelerometer, a three-axis gyroscope, and a three-axis magnetometer. These data streams are then processed as inputs to a linear discriminant analysis classifier, which is used to train the classification model. The overall signal processing workflow for the control strategy is depicted in Fig. 4.

1) Musculoskeletal Model Based on sEMG Signals: The musculoskeletal model simulates the bones, muscles, and joints of the human body, incorporating a priori knowledge of human physiology. It is commonly used to calculate muscle forces, joint moments, and joint angles. One of the most widely recognized models is the Hill-type model, where each muscle is represented by a contractile element, a parallel elastic element, and series elastic element [40], [41]. However, the Hill-type model is often computationally intensive due to the large number of parameters involved, which can make model training time-consuming.

To improve the efficiency and adaptability of the model for clinical real-time control applications, Crouch and Huang [16] simplified the Hill-type model by representing the muscle as a brake with only contractile element and parallel elastic element components, omitting the series elastic element, which is typically used for model tendons. This simplified model, with fewer parameters, requires less training time and provides effective muscle activation decoding [16]. Subsequently, we proposed an improved musculoskeletal model that can accommodate multiple users, designed to predict coordinated wrist and MCP joint movements [42].

The input to this musculoskeletal model is the degree of activation for each muscle. After capturing the raw sEMG signals, we preprocess them to eliminate baseline noise and

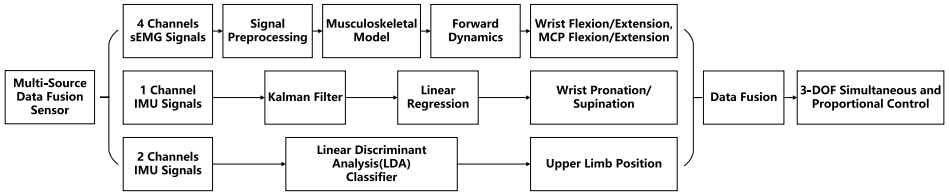


Fig. 4. Signal processing for the control strategy: the four-channel sEMG signals are input into an improved musculoskeletal model to control the flexion and extension of the wrist and MCP joints. An IMU is used to capture forearm movements, with the recorded data enabling control of wrist pronation and supination. The 18-channel data streams from two IMUs are processed as inputs to a linear discriminant analysis classifier, which is trained to classify upper limb positions for effective control.

motion artifacts and normalize the signals. Then both neural and muscle activation are calculated with

$$\mu(t) = \alpha e(t-d) - \beta_1 \mu(t-1) - \beta_2 \mu(t-2) \tag{1}$$

$$a(t) = \frac{e^{\mu(t)} - 1}{e - 1} \tag{2}$$

where $\mu(t)$ is the neural activation at time t, $e(t)$ denotes the normalized EMG signals at time t, d is the delay time, α, β_1, and β_2 are the neural activation coefficients, and $a(t)$ represents the muscle activation at time t.

In the parameter-integrated musculoskeletal model, the force of each muscle is treated as the combined force of the active and passive muscle forces, calculated as

$$F^{\text{total}} = F^{\text{CE}} + F^{\text{PEE}} \tag{3}$$

$$F^{\text{CE}} = f\left(L^{\text{CE}}\right) f\left(v^{\text{CE}}\right) a(t) \tag{4}$$

$$f\left(L^{\text{CE}}\right) = F_0^{\text{CE}}\left(1 - \frac{\left(L^{\text{CE}} - L_0^{\text{CE}}\right)^2}{W\left(L_0^{\text{CE}}\right)^2}\right) \tag{5}$$

$$F^{\text{PEE}} = K^{\text{PEE}}\left(L^{\text{CE}} - L_0^{\text{CE}}\right)^2 \tag{6}$$

where F^{total} is the total muscle force, F^{CE} is the active muscle force generated by the contractile element, F^{PEE} is the parallel elastic element generated by the passive muscle force, L^{CE} is the muscle length, v^{CE} is the muscle contraction velocity, F_0^{CE} is the maximum isometric contractile element force, L_0^{CE} is the optimal contractile element length, and K^{PEE} is the passive elastic element stiffness.

Each muscle has a moment arm for each movement, and the total moment for each movement is the sum of the moments of all the muscles involved, which is calculated as

$$M_i = \Sigma_{j=1}^{4} F_j^{\text{total}} A_{ij} \tag{7}$$

where M_i is the total moment for the ith movement, F_j^{total} is the total force of the jth muscle, and A_{ij} is the force arm of the jth muscle for the ith movement. Specifically, the maximum value of i is set as 2, since the musculoskeletal model is used to decode the two movements of wrist flexion/extension and MCP flexion/extension. Moreover, the force arm of the extensor carpi radialis and the flexor carpi radialis at the MCP joint is set to 0, since these muscles do not pass through this joint anatomically.

Finally, we equate the wrist and palm fingers of the human body as rigid bodies, with each DOF possessing a separate moment of inertia of rotation, and the angular acceleration of each DOF is calculated as

$$\alpha_i = \frac{M_i}{J_i} \tag{8}$$

where α_i is the angular acceleration of the ith movement and J_i is the rotational inertia of the ith movement. The joint angle is obtained by quadratic integration of the fourth-order Runge–Kutta angular acceleration.

Half of the experimental data are used to optimize the parameters of the model, while the rest of the experimental data are used to evaluate the performance of the model. In the 2-DOF musculoskeletal model, each muscle has six parameters to be trained, namely, optimal contractile element length (L_0^{CE}), maximum isometric element contraction force (F_0^{CE}), moment arm at the MCP flexion/extension and wrist flexion/extension joints (M_i), parallel elastic element stiffness (K^{PEE}), and muscle length at the neutral position ($L_{\theta=0}^{\text{CE}}$). Each parameter is limited to an approximation of physiologic values [16]. We use the GlobalSearch function of MATLAB to optimize the muscle parameters and minimize the weighted sum-of-squares difference between the measured and predicted joint angles.

2) Linear Regression Based on Roll Angle: The angles of the wrist joint during pronation and supination are calculated using a linear regression based on the traverse angle of the forearm IMU. The roll angle can be obtained either through the integration of the gyroscope angular velocity or by the calculation of accelerometer tangent. The former method can better reflect the dynamic results of the system, and the latter method does not have the cumulative error, which is more stable in static conditions. However, due to the noise of the sensors adopted, there is an error between the results of both the methods and the true value. Therefore, the Kalman filtering method is used to fuse the results of the two methods into one state estimator to minimize the expected value of the mean square error between the state estimator and the true value.

The geographic coordinate system and carrier coordinate system are defined for solving the attitude of the IMU. Transformation from the geographic coordinate system to the carrier coordinate system can be completed through the rotation of the heading angle, pitch angle, and roll angle. The rotation matrix of the carrier coordinate system relative to the geographic coordinate system can be deduced as

$$\begin{bmatrix} \dot{\gamma} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \frac{1}{\cos\theta} \begin{bmatrix} \cos\theta & \sin\gamma\sin\theta & \cos\gamma\sin\theta \\ 0 & \cos\theta\cos\gamma & -\sin\gamma\cos\theta \\ 0 & \sin\gamma & \cos\gamma \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} \tag{9}$$

where ω_x, ω_y, and ω_z represent the angular velocity of the rotation around the X-, Y-, Z-axis in the carrier coordinate

system, and γ, θ, and ψ represent the roll angle, pitch angle, and heading angle, respectively.

When the Euler angle of the system at the previous moment is known, we can transform the angular velocity in the carrier coordinate system to the geographic coordinate system. When the sampling time Δt is small enough, the Euler angle can be obtained by

$$\begin{bmatrix} \gamma(t) \\ \theta(t) \\ \psi(t) \end{bmatrix} = \begin{bmatrix} \gamma(t-1) \\ \theta(t-1) \\ \psi(t-1) \end{bmatrix} + \begin{bmatrix} \dot{\gamma} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} \Delta t. \tag{10}$$

The roll angle and pitch angle of the accelerometer can be deduced according to the gravity component on each axis. With the local gravitational acceleration of 1 g, the roll angle and pitch angle can be calculated as

$$\theta = \arcsin\left(-A_x^a\right) \tag{11}$$

$$\gamma = \arctan\left(\frac{A_y^a}{A_z^a}\right) \tag{12}$$

where A_x^a, A_y^a, and A_z^a denote the component of gravity sensed by the accelerometer in the direction of each axis.

Considering the measurement errors of the sensors, we use a Kalman filter to fuse the outputs of the accelerometers and gyroscopes, as obtained in (10) and (12), thus to remove the drift in the angle estimation of each sensor. The final estimate of the roll angle is obtained by

$$\gamma = \gamma_{\text{gyro}} + K_k\left(\gamma_{\text{acc}} - \gamma_{\text{gyro}}\right) \tag{13}$$

where K_k is the gain of Kalman optimal estimation.

After obtaining the transverse roll angle, the linear regression is used to predict the joint angle for wrist pronation and supination, as

$$\phi = K\gamma + b \tag{14}$$

where ϕ is the joint angle of the wrist pronation and supination, and K and b are the coefficients of the linear regression. Half of the experimental data were used to train the parameters of the linear regression, and the other half was used to test the ability of the HMI to predict the joint angle of wrist pronation and supination.

To provide readers with a clearer understanding of our design methods, we have created a table summarizing all the design methods, as shown in Table II.

III. Evaluation and Experiments

A. Subjects

All together 12 able-bodied subjects (age 22–25 years, right hand dominant) were recruited for the current study. Among them, one (male) participated in the experiment for sensor performance evaluation, ten (nine males and one female) participated in the online experiment, one (male) participated in the flexible endoscopic robot control experiment, and six (male) participated in the HMI validation experiment. None of the subjects had known neurological diseases. The experimental protocol was approved by the Ethics Committee of Tianjin University (Approval: TJUE-2021-114, Date of Approval: October 6, 2021) and complied with the Declaration of Helsinki.

B. Sensor System Performance Evaluation

With its hardware and software design, the novel HMI system forms a complete loop of "wearing–acquisition–transmission–reception." To evaluate the sensor system performance, a standardized evaluation index and an experimental paradigm are used to compare the proposed sensor system with commercial devices, mainly involving the signal quality assessment. Since the specific parameters of the IMU adopted (MPU9250) have been described in the previous section, here we only evaluate the signal quality of the sEMG.

1) Evaluation Metrics: We choose the metric of SNR to evaluate time-domain characteristics of sEMG signals and the frequency spectrum to evaluate their frequency-domain characteristics. The SNR is mainly obtained by the rms ratio of the signal to the noise, expressed as

$$\text{SNR} = 20\log_{10}\frac{\text{rms}_{\text{signal}}}{\text{rms}_{\text{noise}}} \tag{15}$$

where SNR denotes the SNR of the device, $\text{rms}_{\text{signal}}$ denotes the rms of the sEMG signal acquired during muscle contraction, and $\text{rms}_{\text{noise}}$ denotes the rms of the device noise. Notably, we take the signals acquired during muscle relaxation as the device noise.

We performed Fourier transform on the acquired sEMG signals to obtain the frequency spectrum, which can well reflect the ability of an HMI system to reproduce the core bandwidth of the sEMG signals and to remove the noise in the noncore bandwidth.

2) Control Group: The DTS-Desktop (Noraxon, AZ) and Trigno (Delsys, MA) devices are selected for the comparative evaluation with the proposed sensor system, since they are currently the dominant commercial devices. The DTS-Desktop uses wet electrodes to attach to the skin surface, while the Trigno Wireless, just like our system, uses dry electrodes.

3) Experimental Setup: For the evaluation experiment, a unified experimental environment shall be designed for our system, as well as the two devices in the control group. As shown in Fig. 5, the experimental setup mainly involves the electrode position and grip strength, since the three devices cannot acquire sEMG signals from the exact same position simultaneously. In the experiment, the subjects are required to maintain grip strength as much as possible, and then sEMG signals are collected separately with the three devices.

A force sensor (TH4805B, Botoelc, Hefei) converts a grip strength of 0–500 N into an analog signal (0–5 V), which is then digitized using a data acquisition card (USB-6002, NI, TX) and transmitted to the computer. The computer receives the signal and generates the grip strength curve in real-time. The procedure of the whole experiment is as follows.

1) The dominant hand of the subject is fixed on the grip strength testing device in its natural state.
2) The subject is required to grip the strength gripper with the MVC of the MCP joint for 5 s, and then rest for 5 s. This operation is repeated three times to make one group of the test. Altogether three groups are performed, with an interval of 1 min between adjacent groups.
3) The maximal grip strength of the three groups is averaged and taken as the MVC of MCP joint, and 50%

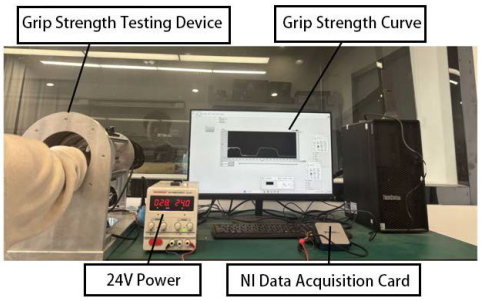


Fig. 5. Experimental setup for sensor system evaluation. A grip strength testing device is used to convert the grip strength of 0–500 N into an analog signal of 0–5 V, which is then converted into a digital signal with a data acquisition card and transmitted it to the computer. The computer receives the signal and generates the grip strength curve in real-time.

of this value is taken as a guideline value for the grip strength, which helps avoid muscle fatigue in the experiment.

4) The subject is asked to wear the three sEMG devices sequentially at the same position to synchronize the acquisition of sEMG signals with grip strength signals and control the force of the MCP joint to keep the grip strength curve around the guideline value. The grip strength is maintained for 5 s, after which a break of 5 s is given. This operation is repeated three times. There is a 5-min interval between the tests with different devices.

C. Online Experiment

As demonstrated in our previous study [42], the online performance of the generic musculoskeletal model surpasses that of the specific model. Therefore, we use a generic musculoskeletal model to construct an online experimental platform for continuous decoding of 3-DOF movements. Instead of the movements of the wrist and MCP joints, a virtual cursor is used to complete the online tasks. The metrics of completion rate, completion time, path efficiency, and the number of overshoots are analyzed to evaluate the online performance of our HMI system based on multisource information fusion.

1) Data Acquisition and Processing: In each trial, sEMG signals and IMU data were collected synchronously. For sEMG signal acquisition, the HMI system recorded data at a sampling rate of 1000 Hz. Prior to electrode placement, the skin surface was cleaned with alcohol to reduce contact impedance. Four sEMG sensors were positioned on four muscles: the extensor carpi radialis longus, extensor digitorum, flexor carpi radialis, and flexor digitorum.

For IMU data collection, an IMU module from the HMI system was affixed to the lateral midsection of the forearm by adhesive tapes. This IMU captured nine data streams (three from the accelerometer, three from the gyroscope, and three from the magnetometer) at a sampling rate of 100 Hz, which were used to decode wrist pronation and supination. In addition, an extra IMU was placed on the lateral upper arm to facilitate for upper limb position classification of HMI.

First, the raw sEMG signals are bandpass-filtered through the fourth-order Butterworth filters in the sensor system, with the cutoff frequencies set at 10 and 500 Hz. Then signals are segmented into consecutive 100-ms windows, and data from the last 20 windows are aggregated. The aggregated sEMG signals are first high-pass-filtered using a first-order Butterworth filter with a cutoff frequency of 40 Hz. The signals are then rectified and subjected to low-pass filtering with a first-order Butterworth filter at a cutoff frequency of 1 Hz. The sEMG data from the second most recent 100 ms window are downsampled to 100 Hz by averaging every ten consecutive data points, and then normalized using the EMG-MVC. Finally, a moving average is calculated over the most recent eight points obtained after downsampling, thus to estimate the activation levels of the four muscles. The accelerometer data from the IMU are low-pass-filtered at 5 Hz, while the remaining data are left unprocessed.

2) Pre-Experimental Model Calibration: Prior to initiating the experiment, each subject stands upright with the dominant-side elbow flexed at 90°, as shown in P1 of Fig. 6. The subject then performs MVC of wrist flexion/extension, MCP flexion/extension, and wrist pronation/supination in sequence, which are recorded to determine the EMG-MVC. The recording of EMG-MVC lasts for 20 s. Subsequently, the subject attempts to maneuver a virtual cursor into each of the eight target positions to be tested. Then the linear gain is adjusted manually to appropriately scale the sEMG signals and output joint angles, ensuring that each subject could achieve a neutral posture while relaxed and comfortably reach all the target positions. This gain adjustment process takes 3 min in average. Once finalized, no further recalibration is required during the online experimental phase.

3) Experimental Paradigm: After the calibration, the subjects are asked to control the black virtual cursor representing the real-time joint angle to reach the gray virtual cursor representing the target joint angle. Inspired by the pseudo-3-D task interface described by Scheme and Englehart [43] and Smith et al. [44], here we use airing virtual cursor to map 3-DOF movements, as illustrated in Fig. 7. The flexion and extension of the wrist joint correspond to the left and right shifts of the cursor, respectively. The flexion and extension of the MCP joint correspond to the contraction and enlargement of the cursor, respectively. The pronation and supination of the wrist joint correspond to the descending and ascending of the cursor, respectively.

The subjects are required to complete online tasks in four limb postures [shown in Fig. 5(a)], to validate whether the HMI system could be applied to movements in daily life. For the online experiment, each subject is required to perform one familiarization trial and two test trials in each upper limb posture, with a 2-min interval between two adjacent trials. In each trial, the subjects need to complete six random target tasks. Notably, the sequence of the upper limb postures is randomized to minimize the influence of muscle fatigue and the subject's training level on the experimental results.

For each trial, the subjects are deemed to get ready when the virtual cursor is maintained within ±8° of the neutral joint angle for 0.5 s, and then the virtual cursor's movement path is recorded. When the cursor is maintained within ±5° of the target joint angle for 0.5 s, it means that the target task is completed. Then the data recording at this target task is ended

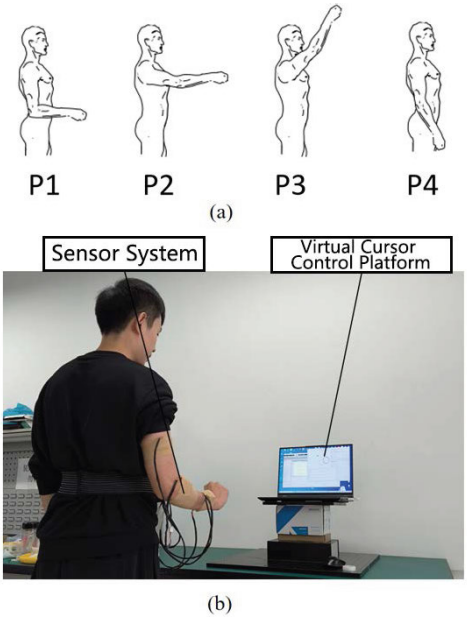


Fig. 6. Setup for the online experiment. The subjects are asked to control the black virtual cursor representing the real-time joint angle to reach the gray virtual cursor representing the target joint angle in four upper limb postures. (a) Four different upper limb postures. (b) Picture of the experimental process.

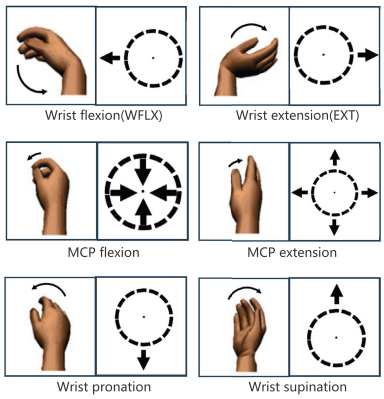


Fig. 7. Virtual cursor operation instructions. The flexion and extension of the wrist joint correspond to the left and right shifts of the cursor, respectively. The flexion and extension of the MCP joint correspond to the contraction and enlargement of the cursor, respectively. The pronation and supination of the wrist joint correspond to the descending and ascending of the cursor, respectively.

and the data are stored. If a subject does not complete the target task within 40 s, the task fails.

D. Flexible Endoscopic Robot Control Experiment

This experiment is performed on a self-developed master–slave flexible endoscopic robot. The movements of the flexible endoscopic robot consists of 3 DOFs (bending, rotation, and translation) [45], which can correspond to the 3 DOFs of our HMI system: wrist flexion/extension, wrist pronation/supination, and MCP flexion/extension, as shown in Fig. 8. This allows our HMI system based on multisource data fusion to properly control the flexible endoscopic robot, presenting a more dexterous device and more intuitive control.

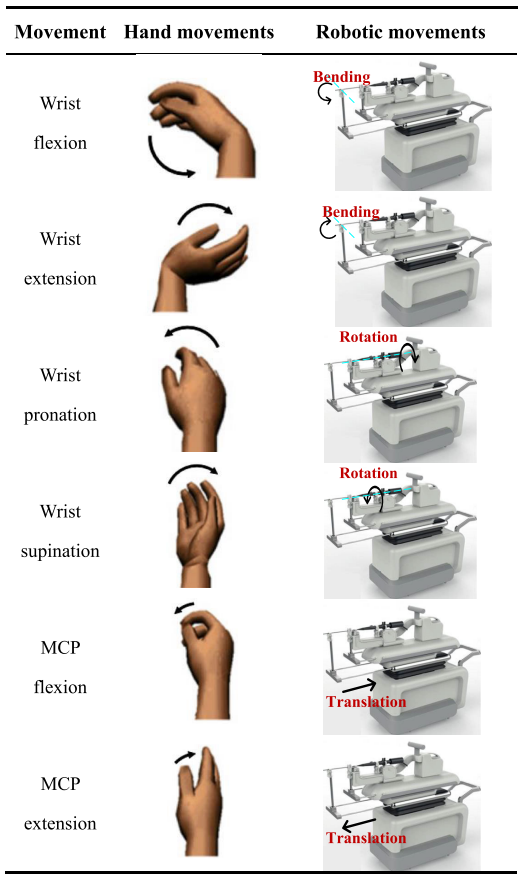


Fig. 8. Mapping of flexible endoscopic robot's DOFs to HMI's DOFs. The movements of the flexible endoscopic robot consists of 3 DOFs (bending, rotation, and translation), which can correspond to the 3 DOFs of our HMI system: wrist flexion/extension, wrist pronation/supination, and MCP flexion/extension.

A bronchoscope (BF-1TQ290, Olympus, Japan) is applied in the control experiment of the flexible endoscopic robot, and a bronchus model fabricated 3-D-printed based on a patient's CT data is used to simulate the actual bronchus, as shown in Fig. 9. The four-channel sEMG data and IMU data of the forearm are collected and the joint angles at the 3 DOFs of MCP flexion/extension, wrist pronation/supination, and wrist flexion/extension are decoded with the proposed HMI system. The joint angles are transformed into three-axis motor revolutions of the PMAC (OMRON, Japan) through mapping, which are filtered by exponential smoothing. Then control of the motor revolutions is implemented: when the wrist pronation/supination or MCP flexion/extension reaches the maximum angle, the robot maintains the current position, and then the subject returns to the neutral position and controls the robot to move cumulatively from the last position. The processed motor revolutions are converted into QString format and sent to an intermediate server using user datagram protocol, which in turn sends them to a computer that is in the same local area network with the flexible endoscopic robot, thus enabling remote control.

E. HMI Validation Experiment

Although real-life experiments have been set up and performed, a statistically relevant and robust approach is needed

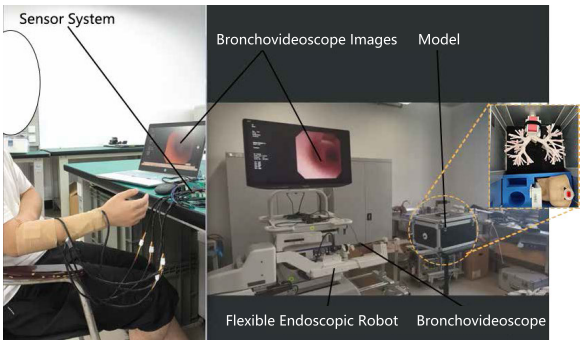


Fig. 9. Schematic of the remote control experiment, with a straight-line distance of 24.2 km between the subject and the flexible endoscopic robot, sharing the bronchoscope images via Tencent Meeting. A bronchoscope (BF-1TQ290, Olympus, Japan) is applied in the control experiment, and a bronchus model fabricated 3-D-printed based on a patient's CT data is used to simulate the actual bronchus.

to validate and evaluate the implemented HMI system. Therefore, six able-bodied participants were recruited to participate in the HMI validation experiment.

1) Data Acquisition and Processing: The acquisition process for sEMG signals and IMU data is consistent with the procedure used in the online experiment, as descried in Section III.

As shown in Fig. 10, the kinematic data of the limb (specifically, the angles of the wrist and MCP joints) were collected using an OptiTrack motion capture system (NaturalPoint, Inc., OR) at a sampling rate of 100 Hz. Fourteen markers were placed on the following anatomical landmarks: the proximal interphalangeal joints of the index and little fingers, the MCP joints of the index and little fingers, the radial styloid, ulnar styloid, mid-radial shaft, medial epicondyle, lateral epicondyle, mid-humeral shaft, medial clavicle, acromion, C7 spinous process, and xiphoid process.

Notably, the motion capture system is equipped with a synchronized output interface. During data acquisition, a step signal is generated, which terminates at the end of data collection. This synchronized output interface is connected to the synchronization interface of the HMI system, ensuring alignment and synchronization between the two systems.

The raw sEMG signals were processed using a fourth-order Butterworth bandpass filter with cutoff frequencies of 10 and 500 Hz in the sensor hardware circuitry. The signals were then full-wave rectified and further processed with a fourth-order Butterworth low-pass filter with a cutoff frequency of 5 Hz. To match the frequency of the kinematic data from the motion capture system, the sEMG signals were downsampled to 100 Hz using a moving rms filter with a window length of 200 samples and a step size of 10 samples. Finally, the processed sEMG signals were normalized to a range of 0–1.

For the IMU data, the accelerometer signals were subjected to a low-pass filter with a 5-Hz cutoff frequency within the MPU9250 hardware, while the remaining IMU data were left unprocessed.

For the kinematic data, joint angles of the upper limb were obtained by fitting the captured motion data to a right-arm model in the biomechanics simulation software OpenSim. The

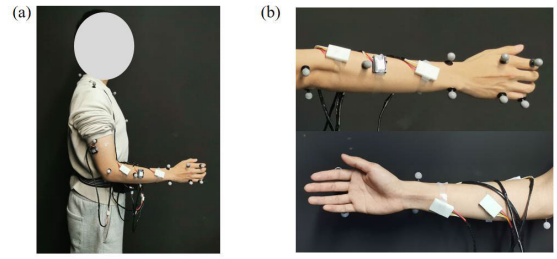


Fig. 10. Setup for the validation experiment. (a) Placement of markers for the Optitrack motion capture system. (b) Placement of sEMG sensors and IMUs.

joint angles of wrist flexion/extension, MCP flexion/extension, and wrist pronation/supination were used as the motion measurements in this study.

2) Experimental Paradigm: The participants were instructed to maintain a standing posture with their torso upright and their elbows bent by 90°. The forearm and hand were fully relaxed, with the fingers naturally flexed toward the palm, as shown in Fig. 10(a). The experiment included seven different types of tasks, comprising rhythmic and random movements.

1) Single-DOF rhythmic wrist flexion/extension movements.
2) Single-DOF rhythmic MCP flexion/extension movements.
3) Single-DOF rhythmic wrist pronation/supination movements.
4) Single-DOF random wrist flexion/extension movements.
5) Single-DOF random MCP flexion/extension movements.
6) Single-DOF random wrist pronation/supination movements.
7) Three-DOF simultaneous random movements.

During the rhythmic tasks, participants performed movements at a fixed frequency of 1/4 Hz. For the random tasks, participants performed movements at a random frequency. Each task type lasted for 20 s, including four trials in total. Each participant completed 28 trials in total. To prevent muscle fatigue, a break of approximately 1 min was provided between consecutive trials.

3) Evaluation Metrics: To evaluate the performance of the controller, the Pearson correlation coefficient (r) and the NRMSE were calculated between the predicted and measured joint angles. The r value reflects the ability of the HMI system to predict the trend of joint angle variations, while the NRMSE value indicates the accuracy in predicting the magnitude of joint angles.

It is important to note that the r and NRMSE values for single-DOF movements were computed exclusively for the corresponding moving joint

$$r = \frac{\Sigma_{i=1}^{n}\left(x_i - \bar{x}\right)}{\sqrt{\Sigma_{i=1}^{n}\left(x_i - \bar{x}\right)^2\left(y_i - \bar{y}\right)^2}} \tag{16}$$

$$\text{NRMSE} = \frac{\sqrt{\frac{1}{n}\Sigma_{i=1}^{n}(x_i - y_i)^2}}{(x_{\max} - x_{\min})} \tag{17}$$

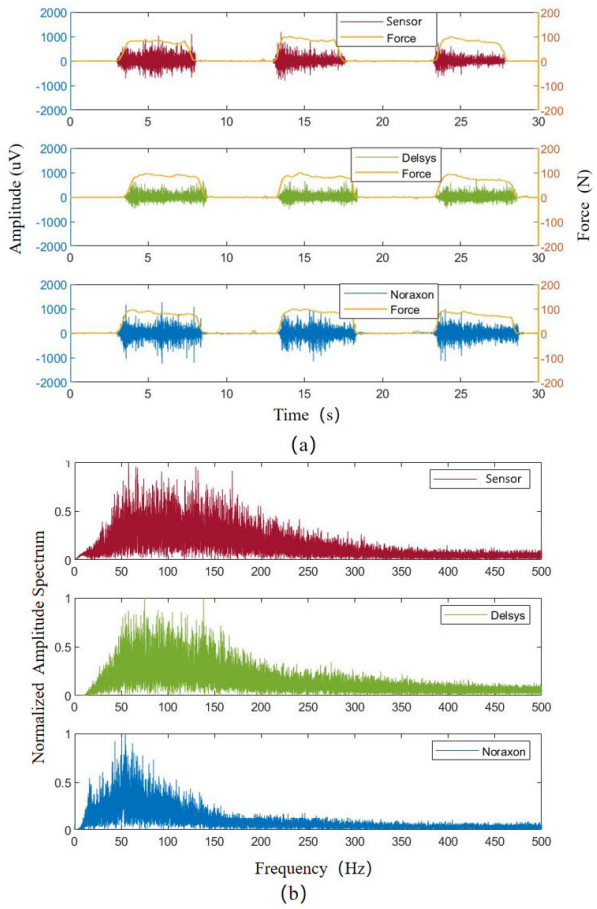


Fig. 11. Time-domain and frequency-domain distribution of sEMG signals obtained by three devices. (a) Time-domain distribution. (b) Frequency-domain distribution.

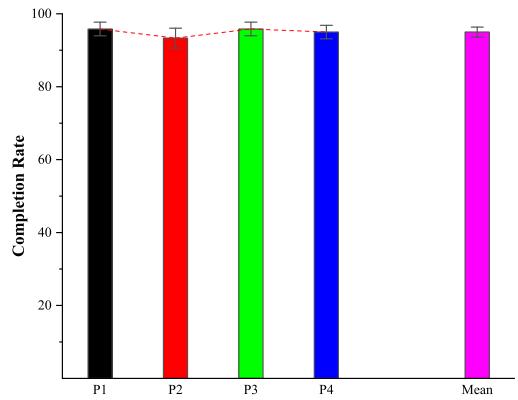


Fig. 12. Average completion rates of ten subjects for each upper limb posture. The rightmost bar is the mean for all the upper limb postures, and the error bars represent the standard error.

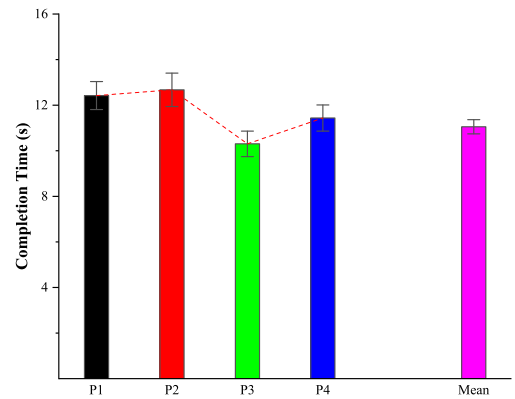


Fig. 13. Average completion time of ten subjects for each upper limb posture. The rightmost bar is the mean for all the upper limb postures, and the error bars represent the standard error.

where x represents the measured joint angle, y represents the joint angle predicted by the HMI, $\bar{x}$ represents the mean of the measured joint angles, $\bar{y}$ represents the mean of the predicted joint angles by the interface, $x_{\max}$ represents the maximum measured joint angle, $x_{\min}$ represents the minimum measured joint angle, and n denotes the data length.

IV. Results

A. Sensor System Performance Evaluation Results

The results for the sensor system performance evaluation are presented in Fig. 11, mainly involving the time-domain characteristics and frequency-domain characteristics of the observed signals. As indicated in Fig. 11(a), the grip strength curves of the three devices are basically the same, so it can be approximated that sEMG signals obtained by the three devices are from the same grip strength. The SNR with the three devices is calculated as 32.54 dB (Delsys Trigno), 33.60 dB (Noraxon DTS-Desktop), and 35.9 dB (our sensor system). According to Fig. 11(b), the sEMG signals from all the three devices predominantly fall within 20–200 Hz.

B. Online Experimental Results

The task completion rates for all the subjects in the four upper limb postures P1, P2, P3, and P4 are 95.833%, 93.333%, 95.833%, and 95%, with an average task completion rate of 95.675%, as shown in Fig. 12.

Fig. 13 demonstrates the mean completion time for tasks completed in all the upper limb postures, which is 12.42217, 12.67393, 10.30391, and 11.43737 s for P1, P2, P3, and P4, respectively, with an average completion time of 11.05337 s.

Fig. 14 illustrates the average path efficiency for completing the tasks in all the upper limb postures, which is 0.14618, 0.15254, 0.158, and 0.12566 for P1, P2, P3, and P4, respectively, with an average value of 0.13831.

Fig. 15 demonstrates the average number of overshoots for completing the tasks at all the upper limb postures, which is 2.73913, 2.57143, 2.12174, and 3.38596 for P1, P2, P3, and P4, respectively, with an average value of 2.56875.

C. Flexible Endoscopic Robot Control Experiment Results

The control experiment starts when the bronchoscope reaches the trachea of the bronchus and ends when it reaches tertiary bronchi, where a notch is set as an endpoint marker, as shown in Fig. 16. The completion time of this operation is recorded and analyzed. The starting time is recorded as 15:35:32 CST, and the end time is 15:37:40 CST, altogether taking 2 min and 9 s (128 s).

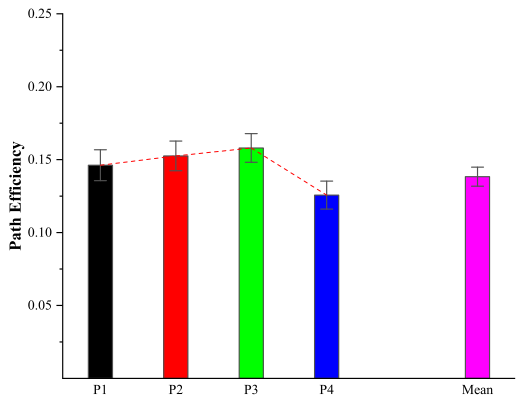


Fig. 14. Average path efficiency of ten subjects for each upper limb posture. The rightmost bar is the mean for all the upper limb postures. Error bars represent standard error.

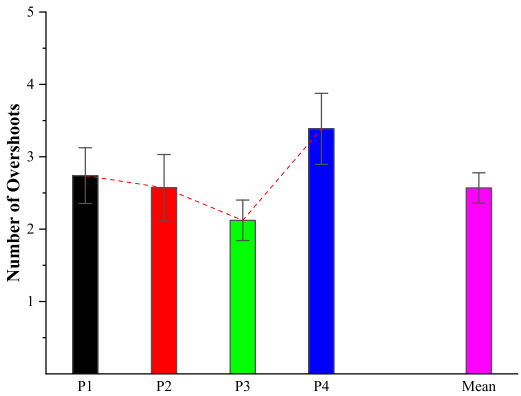


Fig. 15. Average number of overshoots of ten subjects for each upper limb posture. The rightmost bar is the mean for all the supper limb postures. Error bars represent standard error.

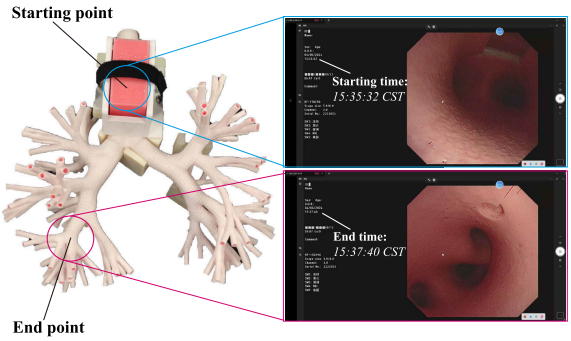


Fig. 16. Flexible endoscopic robot control experiment results. The completion time of this operation is recorded and analyzed. The starting time is recorded as 15:35:32 CST, and the end time is 15:37:40 CST, altogether taking 2 min and 9 s (128 s).

D. Results of the Validation Experiments

Fig. 17 illustrates the r values between the measured and predicted joint angles during various types of movements. For single-DOF rhythmic movements, the average r value between the measured and predicted joint angles was 0.88, 0.94, and 0.98 for MCP flexion/extension, wrist flexion/extension, and wrist pronation/supination, respectively. For the single-DOF random movements, the average r value was 0.87, 0.93, and 0.98, respectively. For the three-DOF simultaneous random movements, the average r value was 0.86, 0.88, and 0.94, respectively. The overall average r value across all types of movements was 0.87, 0.92, and 0.97, respectively.

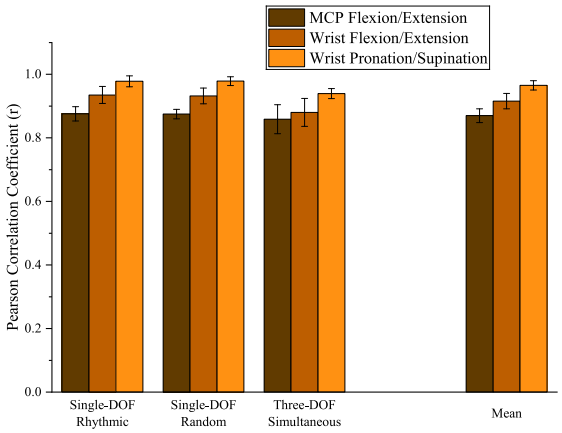


Fig. 17. Pearson correlation coefficient between the predicted and measured joint angles. Error bars represent standard deviation.

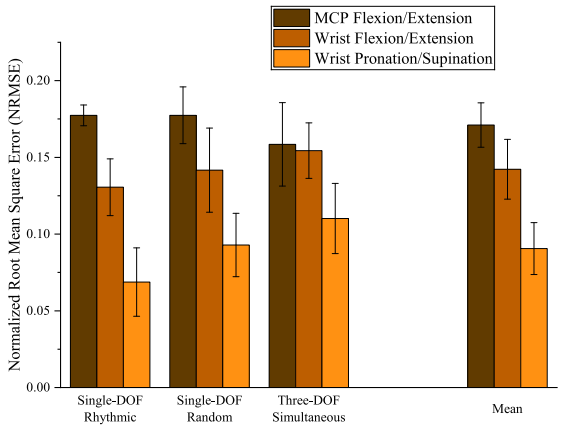


Fig. 18. NRMSE between the predicted and measured joint angles. Error bars represent standard deviation.

Fig. 18 illustrates the NRMSE values between the measured and predicted joint angles during various types of movements. For single-DOF rhythmic movements, the average NRMSE value between the measured and predicted joint angles was 0.18, 0.13, and 0.07 for MCP flexion/extension, wrist flexion/extension, and wrist pronation/supination, respectively. For the single-DOF random movements, the average NRMSE value was 0.18, 0.14, and 0.09, respectively. For the three-DOF simultaneous random movements, the average NRMSE value was 0.16, 0.15, and 0.11, respectively. The overall average NRMSE value across all types of movements was 0.17, 0.14, and 0.09, respectively.

Fig. 19 illustrates the fitting curves of the measured and predicted joint angles during single-DOF random movements and 3-DOF simultaneous random movements. The fitting results indicate that the proposed HMI system can effectively estimate the wrist and MCP joints' movements during both single-DOF and three-DOF simultaneous random movements.

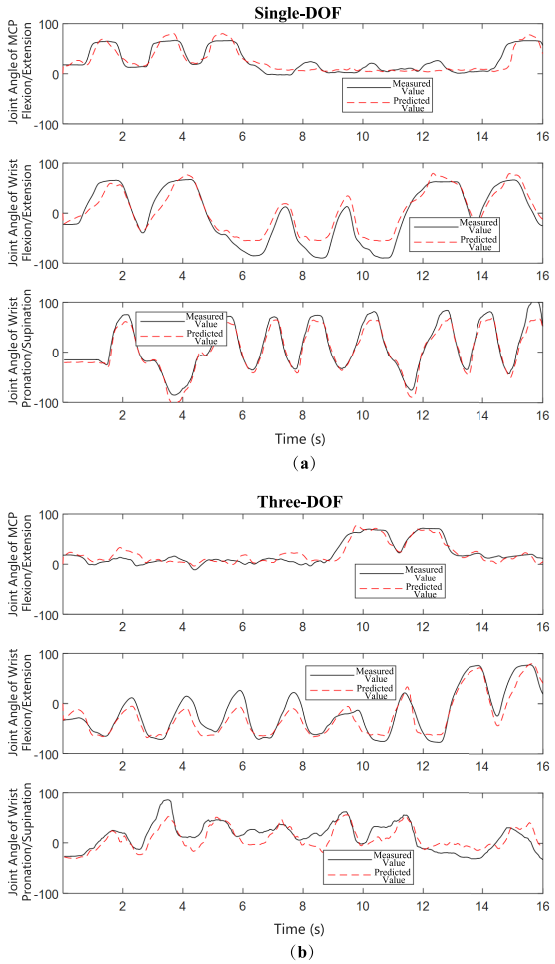


Fig. 19. Measured and predicted joint angles during different types of movements. (a) Single-DOF random movements. (b) Three-DOF simultaneous random movements.

V. Discussion

Multisource data fusion is a solution proposed to address the crosstalk problem encountered by the existing HMI systems based on sEMG signals in decoding multi-DOF movements. In the current study, we use IMU to provide supplementary information for sEMG, thus to overcome the limitation of a single signal source and design a novel HMI system integrating the functions of sensing, control, and actuation. The feasibility of the proposed HMI system in the field of multi-DOF SPC is systematically analyzed. As revealed by the sensor evaluation experiment, our sensor system possesses a higher SNR than Delsys Trigno and Noraxon DTS-Desktop, which can better reproduce sEMG signals and filter sEMG noise. It is worth noting that our sensor system is mainly suitable for decoding upper limb motion intention in multiple DOFs, with a much lower price than that of the two commercial devices. This fully demonstrates that our sensor system can realize the acquisition of sEMG signals and the perception of motion at low cost and high fidelity. Therefore, the proposed HMI system can greatly reduce the cost for low-density sEMG signal acquisition and provide a solid foundation for a wide range of applications.

Impressively, the novel HMI system achieves a completion rate of more than 95% for 3-DOF movements, with low completion time, high path efficiency, and limited overshoots. The musculoskeletal model adopted incorporates a priori knowledge of human physiology to capture the generalized relationship between neuromuscular control signals and human limb motion, and the linear regression based on the IMU roll angle contains the mapping relationship between forearm rotation and wrist pronation and supination rotation angles. Both contribute to the superior completion rate of our HMI system. Moreover, the proposed HMI system can realize accurate and stable SPC, whose feasibility and practicality in four common upper limb postures are verified through online experiment. In addition, it is interesting to observe that P3 among the four positions witnesses the best online control performance, where the muscle is tensed and the sEMG signals are more pronounced.

In addition, the results of the flexible endoscopic robot remote control experiment show that it is feasible to use the fusion of sEMG and IMU as the control input to the HMI system, with good and stable SPC capability. This provides a new direction for surgeons to get rid of traditional control methods and realize intuitive and easy control of the surgical robot.

Although the proposed HMI is applied to actual robotic control, we conducted an additional statistical validation experiment for the system. The control performance of the HMI was quantified by calculating r and NRMSE values between the actual joint angles measured with the Optitrack motion capture system and the joint angles predicted by the HMI system. The r value effectively reflects the ability of the HMI system to predict the trend of joint angle variations, and a value closer to 1 indicates better performance in trend prediction. The NRMSE value reflects the ability of our system to predict the amplitude of joint angles, and a value closer to 0 indicates a smaller prediction error for amplitude. The experimental results demonstrate that the proposed HMI can reliably estimate wrist and MCP joints movements during both single-DOF and three-DOF movements.

The current study still has some limitations. We only attempted to predict the joint angle of wrist pronation and supination rotation by linear regression of the rotation angle of the forearm. In our future work, different data streams and models will be applied to predict the joint angles. Moreover, no wrist amputees were involved in the current study, and we plan to recruit amputees to better demonstrate the applicability of our HMI system. Furthermore, the HMI proposed in this study is currently limited to applications on the wrist and MCP joints. For use on other body parts, such as an arm or a leg, sEMG sensors can be placed on the corresponding antagonistic muscles to capture sEMG signals, while IMUs can be positioned on the target limb to track its movements. The DOFs of the associated joints can be accurately decoded by training the corresponding musculoskeletal models. This approach represents a promising direction for future research. In addition, future research may explore the transition toward information fusion. By integrating more advanced features, information fusion can enhance the robustness and adaptability

of HMI systems. Potential directions include incorporating other sensor modalities such as tactile or visual feedback.

VI. Conclusion

This article proposes a novel HMI system-based multisource information fusion and systematically analyzes and demonstrates the feasibility of integrating sEMG and IMU in the field of multi-DOF SPC. The designed sensor system collects sEMG signals with the SNR of 35.9 dB, which is much lower in cost than commercial devices, with closer time-domain and frequency-domain characteristics. In the online experiment, the average task completion rate for ten subjects is as high as 95%. The remote SPC of the flexible endoscopic robot is realized, and the proposed HMI system shows online practicality as well as reliable and stable SPC capabilities. This research lays a good foundation for the further development of the HMI system with multisource information fusion.

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